



UNIVERSITY^{AT}ALBANY
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CSI 436/536 (Spring 2026)
Machine Learning

Lecture 12: Probabilistic Models

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Mar 25, 2026

Today

- Lectures after midterm
- Discriminative model vs generative model
- Maximum likelihood estimation
 - Linear regression

Lectures after midterm

- Lecture 12: Probabilistic models in supervised learning
 - Maximum likelihood estimation
 - Naïve Bayes model
- Lecture 13-16: Advanced techniques to improve supervised learning
 - Theoretical foundation: Error decomposition
 - Ensemble method (combining multiple classifiers)
 - Kernel method (feature transformation)
 - Non-linear model (neural network)
- Lecture 17-18: Unsupervised learning
 - Clustering
 - Dimension reduction
- Lecture 19: Advanced topic: Decision making
- Lecture 20: Course review

So far we have learned a lot about ML, but...

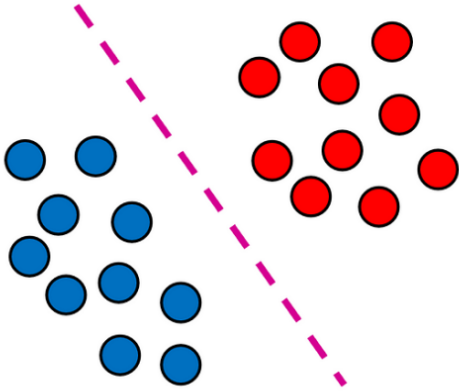
- We learned how to
 - Specify a hypothesis class
 - Work out the possible shapes of decision boundaries
 - Train a model by solving an optimization problem
- How did we come up with the hypothesis classes in the first place?
 - We brainstormed... and used
 1. decision trees
 2. linear-classifiers, thresholding a weighted linear combination of features.
 - But how do we know the resulting decision boundaries are **appropriate** for the problems we hope to solve?

We learned about directly modelling the predictive functions. There is another way... called “Probabilistic modelling”

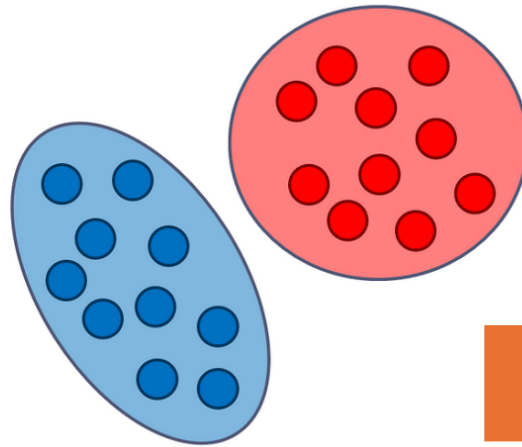
- We can model how the data is generated in the first place.
 - Model the labeling process via a **conditional distribution** $P(y|x)$.
 - ***(Probabilistic) discriminative model***
 - Non-probabilistic discriminative models:
 - Decision-trees
 - Linear classifiers
 - Model the **joint distribution** $P(x, y)$ by modeling the label distribution $P(y)$ and a generative process $P(x|y)$.
 - ***Generative model.***

Discriminative models vs generative models

Discriminative



Generative



	Non-probabilistic	Probabilistic
Discriminative model (how data can be separated?)	Modeling predictive function	Modeling $P(y x)$
Generative model (How data is generated?)		Modeling $P(x, y)$ by label distribution $P(y)$ and generative process $P(x y)$

Probabilistic modelling

- Hard prediction:
 - $h(x) = \operatorname{argmax}_y P(y|x)$
 - “Bayes optimal”:
 - If the label generative process is indeed $P(y|x)$
- Soft prediction $P(y|x)$
 - Quantifying uncertainty
 - More informative than the score function
 - More interpretable / explainable

How to model $P(y|x)$?

- Bayes rule:
 - $P(y|x) = \frac{P(y)P(x|y)}{P(x)}$
- Key idea:
 - $\operatorname{argmax}_y P(y|x) \leftrightarrow \operatorname{argmax}_y P(y)P(x|y)$
 - Why?
 - y doesn't depend on $P(x)$.
 - $P(y)$ (distribution of label):
 - Can be estimated by counting labels in training set.
 - $P(x|y)$ (data generating process)

Directly modeling $P(x|y)$ is challenging

Binary vectors, 2^3 rows +
binary output $Y \in \{0, 1\}$

x_1	x_2	x_3
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

1. What is the number of data points needed to estimate $p(x|y)$?

2. What happens if there are d binary features?

We need maximum likelihood estimation!

Maximum likelihood estimation

- Used since Gauss, Laplace, etc....
- Popularized / carefully analyzed by Ronald Fisher.
- Which distribution is more *likely* to have produced the data?

$$\max_{P \in \Pi} f_{\text{Data} \sim P}(\text{Data})$$

What is the difference between **probability** and **likelihood**?

- $P(\text{Data}; \text{Parameter})$
 - If it is a function of the data, then it's probability.
 - If it is a function of the parameter while the data is fixed, then it is likelihood.

Recap: Estimating the mean of Gaussian distribution

- Data

$$X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2)$$

- Likelihood:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right)$$

- The MLE problem:

$$\hat{\mu} = \arg \max_{\mu \in [0,1]} \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(X_i - \mu)^2}{2\sigma^2}}$$

MLE for linear regression

- $P(y|x)$ is modeled by “Linear Gaussian model”

$$y_i = x_i^T \theta^* + \epsilon_i \quad \text{where } \epsilon_i \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.}$$

- Data:

$$(x_1, y_1), \dots, (x_n, y_n)$$

- In-class exercise: Work out the optimization problem to solve for the MLE for θ^* .